Third Year (5th Semester)

Theory

Theory of Computation

Subject Code:		**Total Contact Hour**	**30**
Semester:	5TH	**Total Credit**	**3**
Subject Name:	**Theory of Computation**		
Pre-requisites (if any):	**Data structures, Algorithm Design, Discrete Mathematics**		
Course Objectives:	1. To give an overview of the theoretical foundations of computer science from the perspective of formal languages 2. To illustrate finite state machines to solve problems in computing 3. To familiarize Regular expression, Regular grammars, 4. To familiarize context frees grammar. 5. To explain the hierarchy of problems arising in the computer sciences.		

SYLLABUS		
Module I	Introduction to Finite Automata (FA), Deterministic Finite Automata (DFA) -Formal definition, simpler notations (state transition diagram, transition table), language of a DFA. Nondeterministic Finite Automata (NFA)- Definition of NFA, language of an NFA, Equivalence of Deterministic and Nondeterministic Finite Automata, Applications of Finite Automata, Finite Automata with Epsilon Transitions, Eliminating Epsilon transitions, Minimization of Deterministic Finite Automata, Finite automata with output (Moore and Mealy machines) and Inter conversion.	**6 Hours**
Module II	Introduction to Regular Expression, Identities of Regular Expressions, Finite Automata and Regular Expressions- Converting from DFA's to Regular Expressions, Converting Regular Expressions to Automata, applications of Regular Expressions. Regular Grammars Definition, regular grammars and FA, FA for regular grammar, Regular grammar for FA. Proving languages to be non-regular -Pumping lemma, applications, Closure properties of regular languages.	**6 Hours**
Module III	Context-free grammar and Derivation Trees, Sentential Forms, Rightmost and Leftmost derivations of Strings. Ambiguity in CFG's, Minimization of CFG's, CNF, GNF, Pumping Lemma for CFL's, Enumeration of Properties of CFL.	**6 Hours**
Module IV	Pushdown Automata and Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA. Turing Machines and Formal definition and behaviour, Languages of a TM, TM as accepters, and TM as a computer of integer functions, Types of TMs.	**6 Hours**
Module V	Recursive And Recursively Enumerable Languages (REL): Properties of recursive and recursively enumerable languages, Universal Turing machine, The Halting problem, Undecidable problems about TMs. Context sensitive language and linear bounded automata (LBA), Chomsky hierarchy, Decidability, Post's correspondence problem (PCP), undecidability of PCP.	**6 Hours**
Essential Reading	1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman (2007), Introduction to Automata Theory Languages andComputation, 3rdedition, Pearson Education, India. 2. Elements of the Theory of Computation, 2nd edition. Harry Lewis, Christos Papadimitriou, Prentice Hall.	
Supplementary Reading	1. K. L. P Mishra, N. Chandrashekaran (2003), Theory of Computer Science-Automata Languages and Computation, 2nd edition, Prentice Hall of India, India.	

	2. Introduction to the Theory of Computation, 3rd edition. Michael Sipser, Cengage Publications 3. Automata and Computability, Dexter C. Kozen. Part of the Undergraduate Texts in Computer Science book series (UTCS), Springer.
Course Outcomes:	CO1: To use basic concepts of formal languages of finite automata techniques CO2: To Design Finite Automata's for different Regular Expressions and Languages CO3: To Construct context free grammar for various languages CO4: To solve various problems of applying normal form techniques, push down automata and Turing Machines CO5: To understand the computational limitations of a given problem.

Operating Systems

Subject Code:	PC	**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Operating Systems**		
Pre-requisites (if any):			
Course Objectives:	1. The objective of the course is to analyze the concepts of Operating System and process. 2.Illustrate the Scheduling of a processor for a given problem instance, 3. Identify the dead lock situation and provide appropriate solution 4.Analyze memory management techniques and implement page replacement Algorithm 5.Understand the implementation of file systems and directories		

SYLLABUS		
Module I	**Introduction**: What is and Operating Systems. Simple Batch Systems, Multiprogramming and Time-Sharing systems. Personal Computer Systems, Parallel Systems, Distributed systems and Real time Systems. Operating Systems structures: systems components, protection system, O.S. Services, system calls. **Processes**: Introduction, Process states, process management, Interrupts, Interprocess Communication Threads: Introduction, Thread states, Thread Operation, Threading Models. **Processor Scheduling**: Scheduling levels, pre-emptive vs no pre-emptive scheduling, priorities, scheduling objective, scheduling criteria, scheduling algorithms, demand scheduling, real-time scheduling.	**8 Hours**
Module II	**Process Synchronization**: Mutual exclusion, software solution to Mutual exclusion problem, hardware solution to Mutual exclusion problem, semaphores, Critical section problems. Case study on Dining philosopher problem, Barbershop problem etc. **Deadlocks**: Examples of deadlock, resource concepts, necessary conditions for deadlock, deadlock solution, deadlock prevention, deadlock avoidance with Bankers algorithms, deadlock detection, deadlock recovery.	**7 Hours**
Module III	**Memory Organization & Management**: Memory Organization, Memory Hierarchy, Memory Management Strategies, Contiguous versus non-Contiguous memory allocation, Partition Management Techniques, Logical versus Physical Address space, swapping, Paging, Allocation of frames, Segmentation, Segmentation with Paging **Virtual Memory**: Demand Paging, Page Replacement, Page-replacement Algorithms, Performance of Demand Paging, Thrashing, Demand Segmentation and Overlay Concepts.	**7 Hours**
Module IV	**File System**: Introduction, File Organization, Logical File System, Physical File System, File Allocation strategy, Free Space Management, File Access Control, Data Access Techniques, Data Integrity Protection, Case study on file system viz FAT32, NTFS, Ext2/Ext3 etc.	**4 Hours**

Module V	**I/O Systems** Kernel I/O Subsystem, Disk Structure, Disk Scheduling, Disk Management, Swap Space Management, RAID Structure, Case study on Linux System, Case study on Windows XP.	**4 Hours**
Essential Reading	1.Operating System Concepts: Abraham Silberschatz and Peter Bear Galvin, Addison Wesley 2. Operating System, Madnik and Donovan, McGraw Hill,.	
Supplementary Reading	1.Moswen O.S., Andrew, S. Tannenbaum, PHI 2. Operating Systems and system programming, P. Blkeiahn Prasad, SCITECH,	
Course Outcomes:	CO1: Analyze the concepts of Operating System and process. CO2: Illustrate the Scheduling of a processor for a given problem instance. CO3: Identify the dead lock situation and provide appropriate solution. CO4: Analyze memory management techniques and implement page replacement Algorithm. CO5: Understand the implementation of file systems and directories.	

Artificial Intelligence and Machine Learning

Subject Code:		**Total Contact Hour**	**30**
Semester:	4th / 5th	**Total Credit**	**3**
Subject Name:	**Artificial Intelligence and Machine Learning**		
Pre-requisites (if any):			
Course Objectives:	1. To familiarize students with the fundamental concepts, theories, and applications of Artificial intelligence& Machine learning. Students will gain insight into the various subfields of AI& ML. 2. Students will have a clear understanding of the fundamental concepts and terminology of Artificial intelligence& Machine learning, enabling them to discuss and comprehend AI-related topics. 3. Students will have a clear understanding about neural networks, Fuzzy logic. 4. Students will have a clear understanding about Clustering and related techniques. 5. Students will have a clear understanding about Classification and related techniques.		
SYLLABUS			
Module I	Introduction to Artificial Intelligence, Applications of AI, State-space problem, Problem solving by Intelligent search: BFE, DFS, Iterative Deepening Search, Hill climbing, Heuristic search: A*, AO*, MIN_MAX Algorithm, Alpha-beta cutoff		**8 Hours**
Module II	Knowledge representation and reasoning: Formalized symbolic logic, propositional logic, First-order predicate logic, wff conversion to clausal form, inference rules, resolution principle.		**5 Hours**
Module III	Unsupervised Learning: K-means, K-Medoids, Hierarchical clustering, Density based clustering, Validation Method: LOO, K-fold cross validation.		**5 Hours**
Module IV	Supervised Learning: Decision Tree, Naïve Bayes classifier, K-NN, Introduction to regression. Performance matrix: Confusion matrix, Precision, Recall, Sensitivity, Specificity, MAE, MSE		**6 Hours**
Module V	Neural Network Artificial Neuron and its model, activation functions, Neural network architecture: single layer and multilayer feed forward networks, recurrent networks, Training of ANN, Back propagation, RBFNN.		**6 Hours**
Essential Reading	1.E.Rich and K. Knight, Artificial Intelligence-TMH 2.Neuro Fuzzy and Soft Computing, J. S. R. JANG,C.T. Sun, E. Mitzutani, PHI		
Supplementary	1.Artificial Intelligence, Dan W Patterson, Prentice Hall of India		

Reading	2. Computational Intelligence Principles, Techniques and Applications, Amit Konar, Springer publication. 3. M. Gopal, Applied Machine Learning, McGraw Hill Education
Course Outcomes:	CO1: Understand the basics of Search techniques, Knowledge representation and reasoning in Artificial Intelligence. CO2: Understand the Supervised machine learning and Unsupervised machine learning. CO3: Analyzevarious machine learning models. CO4: Implement various Supervised machine learning techniques and analyze them. CO5: Implement various Unsupervised machine learning techniques and analyze them.

Professional Elective - I	
Specialization-1	Computational Intelligence
Specialization-2	Data Mining and Data Warehousing
Specialization-3	Microprocessors and Microcontrollers
Specialization-4	Distributed Systems
Specialization-5	Cryptographic Foundation and Network Security
Specialization-6	Object-Oriented Analysis and Design
Specialization-7	Web Technology

Computational Intelligence

Subject Code		Total Contact Hour	30
Semester		Total Credit	03
Subject Name	**Computational Intelligence**		
Pre-requisites (if any)	Basic Mathematical Knowledge		
Course Objective	1. To understand, analyze and apply the concepts of Neural network, neuro-modelling, several neural network paradigms. 2. To understand, analyze and apply the concepts of fuzzy logic, knowledge representation using fuzzy logic, approximation reasoning, fuzzy inference system, fuzzy logic control and other machine intelligent application of fuzzy logic. 3. To understand, analyze and apply the concept of evolutionary computing paradigm known as genetic algorithm to engineering optimization problems. 4. To understand, analyze and apply the concept of hybrid algorithms in different engineering application		
SYLLABUS			
Module I	**Introduction:** Artificial Intelligence, pitfalls of the Traditional AI, Computational Intelligence-a Formal Definition, soft computing-Definitions, Fundaments elements of soft computing, Computational Learning Theory, synergism in soft computing: Neuro-Fuzzy synergism, Neuro-GA synergism, Fuzzy-GA synergism		04 **Hours**
Module II	**Neural Network :** Artificial Neuron and its model, activation functions, Neural network architecture: single layer and multilayer feed forward networks, recurrent networks, perception model, feed forward neural network, Back propagation, Adaline, Widrow-Hoff's Adaline model, Madaline, Unsupervised learning neural network: Hopfield neural network, Competitive learning, self-organizing feature map, Reinforcement learning: Q-learning, Temporal difference learning.		06 **Hours**
Module III	**Fuzzy Logic :** Basic concepts of fuzzy logic, Fuzzy sets and Crisp sets, membership functions, Fuzzy set theory and operations, Extension principle of fuzzy set, fuzzy inference, Fuzzy implications, fuzzy relation, fuzzy reasoning , fuzzy c-means clustering , fuzzy inference Engine on VLSI architecture, Defuzzification techniques		08 **Hours**
Module IV	**Genetic Algorithm :** Fundamentals of genetic algorithms: Encoding, Fitness functions, Reproduction Genetic Modeling: Cross cover, Inversion and deletion, Mutation operator, Bit-wise operators, Bitwise operators used in GA.		08 **Hours**

	Convergence of Genetic algorithm. GA as an alternative to back propagation, Applications of GA in navigational planning of robots , Real life Problems	
Module V	Hybrid Systems: Neuro-fuzzy synergism, weakly coupled Neuro-fuzzy system, Tightly coupled Neuro-Fuzzy System, fuzzy-GA synergism, Neuro-GA, Adaptation of neural learning algorithm using GA	04 **Hours**
Essential Reading	1. **Computational Intelligence** Principles, Techniques and Applications, Amit Konar, Springer publication. 2. Neural Networks, Fuzzy Logic, and Genetic Algorithm (synthesis and Application) S.Rajasekaran, G.A. VijayalakshmiPai, PHI 3. **Principles of Soft Computing** S. N. Sivanandam and S. N. Deepa, Wiley-India Edition	
Supplementary Reading	1. Neuro Fuzzy and Soft Computing, J. S. R. JANG,C.T. Sun, E. Mitzutani, PHI 2. Soft-computing Fundamentals and Applications, D. K. Pratihar, Alpha Science	
Course Outcomes	1. To know about the basic concept of computational intelligence and also their use in some real life situation 2. To solve the problems using neural network techniques 3. To find the solution using different fuzzy logic techniques 4. To use genetic algorithms for different modelling 5. To integrate the various evolutionary computing.	

Data Mining and Data Warehousing

Subject Code:		**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Data Mining and Data Warehousing**		
Pre-requisites (if any):	Basics of Programming in python, Mathematics and Statistics		
Course Objectives:	1. To understand the principles of Data warehousing and Data Mining. 2. To understand the overall knowledge discovery process in databases. 3. Apply preprocessing techniques to prepare data for mining. 4. To Design Data Mining models for classification, association, and prediction of data. 5. To apply and make use of data mining concepts.		
SYLLABUS			
Module I	Data Mining overview, Data Warehouse and OLAP Technology Data Warehouse Architecture, Steps for the Design and Construction of Data Warehouses, A Three-Tier Data Warehouse Architecture, OLAP, OLAP Queries, Metadata Repository, Data Preprocessing: Data Integration and Transformation, Data Reduction, Data Mining Primitives, System Architectures – Data Mining Primitives: What Defines a Data Mining Task? Task-Relevant Data, The Kind of Knowledge to be Mined, KDD		**3 hours**
Module II	Data Mining: Data Mining Functionalities, Data Preprocessing, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation- Architecture of A Typical Data Mining Systems- Classification of Data Mining Systems. Mining Association Rules in Large Databases, Association Rule Mining, Market Basket Analysis: Association Rule Mining, Basic Concepts, Association Rule Mining-A Road Map, Mining Association Rules from Frequent itemset, Mining Multilevel Association Rules from Transaction Databases, Multilevel Association Rules, Approaches to Mining		**8 hours**

	Multilevel Association Rules, Mining Distance-Based Association Rules, From Association Mining to Correlation Analysis.	
Module III	Classification and Prediction: Issues Regarding Classification and Prediction, Classification by Decision Tree Introduction, Bayesian Classification, Rule Based Classification, Classification by Back propagation, Support Vector Machines, Associative Classification, Lazy Learners, Other Classification Methods, Prediction, Accuracy and Error Measures, Evaluating the Accuracy of a Classifier or Predictor, Ensemble Methods, Model Section.	**10 hours**
Module IV	Cluster Analysis: Types of Data in Cluster Analysis, A Categorization of Major Clustering Methods, Partitioning Methods, Hierarchical methods, Density-Based Methods, Grid-Based Methods, Model-Based Clustering Methods, Clustering High-Dimensional Data, Constraint-Based Cluster Analysis, Outlier Analysis.	**6 hours**
Module V	Mining Object, Spatial, Multimedia, Text and Web Data: Multidimensional Analysis and Descriptive Mining of Complex Data Objects, Spatial Data Mining, Multimedia Data Mining, Text Mining, Mining the World Wide Web.	**3 hours**
Essential Reading	1. Jiawei Han, Micheline Kamber and Jian Pei"Data Mining Concepts and Techniques", Third Edition, Elsevier. 2. M. H. Dunham, " Data Mining Introductory and Advanced Topics", Pear son Education.	
Supplementary Reading	1. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, "Introduction to Data Mining", Pearson Education. 2. Alex Ber son and Stephen J. Smith, "Data Warehousing, Data Mining & OLAP", Tata McGraw Hill Edition, Tenth Reprint. 3. Prabhu, "Data warehousing - concepts, Techniques, Products and Application s", Prentice Hall of India. 4. Richard Royger, Data mining – 1st Edition, CRC Press, 2nd Edition 5. Ian H. Witten et.al, Data Mining: Practical Machine Learning Tools and Techniques, Morgan Kofman publishers. 4th edition	
Course Outcomes:	CO1: Learn Data Mining overview, Data Warehouse and OLAP Technology. CO2: Understand the Architecture of a Data Mining system. CO3: Perform Classification, Association, and Prediction of data. CO4: Analyze and compare various classification models in data mining. CO5: Understand the Applications and Trends in Data Mining.	

Microprocessors and Microcontrollers

Subject Code:		**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Microprocessors and Microcontrollers**		
Pre-requisites (if any):			
Course Objectives:	1. The objective of the course is to understand basic structure of the microprocessors like 8085, 8086 etc, their functions timing diagram & learn assembly language coding for the above processors. 2. Understand the functions of microcontrollers and be able to code, interface the devices using them.		

SYLLABUS		
Module I	8085-architecture, operation, pin configuration and functions, bus organization, control signal generation for external operations- fetch, IO/M, read/write, machine cycles. Addressing mode, instruction set. Assembly language of 8085.	**6 Hours**
Module II	Overview of 8086 microprocessor. Signals and pins of 8086 microprocessor. Assembly language of 8086, Description of Instructions.	**6 Hours**
Module III	8051-architecture, operation, pin configuration and functions, memory organization, register, I/O ports, addressing modes, instruction sets, instruction classification. Assembly language programming Interrupts in 8051.	**6 Hours**
Module IV	Interfacing with keyboards, LEDs, 7 segment LEDs, LCDs, Interfacing with ADCs, Interfacing with DACs, etc.	**6 Hours**
Module V	Introduction to the Arduino/ Raspberry Pi, creating an programming Environment, IDE, Libraries& Programming, Analog and Digital Interfacing, Adding Interrupts, communicating with devices and sensors.	**6 Hours**
Essential Reading	1. Krishna Kant, "Microprocessors and Microcontrollers Architecture, programming and system design using 8085, 8086, 8051 and 8096". PHI 2. Douglas V Hall, "Microprocessors and Interfacing, Programming and Hardware" TMH 3.Ramesh S. Goankar "8085 Microprocessors Architecture Application and Programming", PHI	
Supplementary Reading	1. Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D.MCKinlay The 8051 Microcontroller and Embedded Systems, Second Edition, Pearson Education. 2. Kenneth J. Ayala, "The 8086 Microprocessor: Programming & Interfacing The PC", Delmar Publishers. 3. Michael Margolis,"Arduino Cookbook", , O'Reilly Media, Inc. 4. John Baichtal, "Arduino for beginners: Essential Skills Every Maker Needs", Person Education. 5. Eben Upton and Gareth Halfacree "Raspberry Pi User Guide", 4th Edition, JohnWiley & Sons. 6. Simon Monk, "Programming with Raspberry Pi: Getting Started with Python", McGraw Hill Professional, 2012	
Course Outcomes:	CO1: Understand and distinguish 8085, 8086 architecture and functions CO2: Learn the structure of micro controllers like 8085. CO3: Learn programming using microprocessors & microcontrollers. CO4: Design interfacing devices with microprocessors. CO5: Develop interfacing using Aurdino/ Raspberry Pi.	

Distributed Systems

Subject Code:		**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Distributed Systems**		
Pre-requisites (if any):			
Course Objectives:	1. To learn the principles, architectures, algorithms and programming models used in distributed systems. 2. To understand the heterogeneous systems such as computers, mobile phones, other devices and Internet) and their functionalities. 3. To design and implement sample distributed systems		

SYLLABUS		
Module I	**Introduction to Distributed systems**: Review of Networks, Operating Systems, Concurrent Programming, Types of Distributes Systems, Goals, System architectures.	**5 Hours**
Module II	**Distributed Computing Paradigms:** Communication in Distributed Systems, Types of communication**.** Basic Message Passing Model – The Client Server, Group communication, Message Passing, RPC basics, RPC implementation, RPC communication and issues **Naming**: Flat naming, Structured naming, Attribute based naming.	**6 Hours**
Module III	**Process models in distributed systems:** Processes, Threads, Virtualization, Clint, Server, Code Migration **Synchronization:** Clock Synchronization, Logical clocks, Distributed Mutual Exclusion, Global positioning of nodes, Election Algorithms.	**6 Hours**
Module IV	**Consistency and Replication:** Motivation, Object Replication, Consistency Models, Consistency Protocols **Fault Tolerance:** Failure Models – Process Resilience – Reliable Client Server and Group Communications – Distributed Commit Protocols – Check-pointing and Recovery - Distributed Databases - Distributed Transactions	**6 Hours**
Module V	**Distributed File System:** File system, DFS- definition, Characteristics, Goals, NFS Architecture, NFS Implementation, Protocols, The CODA file system-Design Overview, Handling Byzantine Failures, High Availability in Peer-to-Peer Systems, Security in NFS, Decentralized Authentication. Distributed Web-Based Systems: Architecture Distributed Coordination-Based Systems: Architecture	**7 Hours**
Essential Reading	1. Andrew S. Tanenbaum, Maarten Van Steen, "Distributed Systems Principles and Paradigms", 2nd ed., Pearson Education. 2. George Coulouris, Jean Dollimore, and Tim Kindberg, "Distributed Systems Concepts and Design", 5th ed., Pearson Education.	
Supplementary Reading	1. P. K. Sinha, Distributed Operating Systems: Concepts and Design, IEEE press 2. M. Singhal and N. G. Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill	
Course Outcomes:	CO1: Gain the knowledge of the basic elements and concepts related to distributed system technologies CO2: Learn about the core architectural aspects of distributed systems CO3: Demonstrate knowledge of details the main underlying components of distributed systems (such as RPC, file systems)	

	CO4: Use and apply important methods in distributed systems to support scalability and fault tolerance CO5: Design and implement large-scale distributed applications

Cryptographic Foundation and Network Security

Subject Code:		**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Cryptographic Foundation and Network Security**		
Pre-requisites (if any):	A strong foundation in mathematics and computer science. Cryptography involves using mathematical concepts and algorithms to create secure communication systems.		
Course Objectives:	1. Understand the threats and security mechanisms for Computer Networks. Identify computer and network security threats, classify the threats and develop a security model to prevent, detect and recover from the attacks. 2. Encrypt and decrypt messages using block ciphers, sign and verify messages using well known signature generation and verification algorithms. 3. To be able to secure a message over insecure channel by various means. 4. To learn about how to maintain the Confidentiality, Integrity and Availability of a data. 5. To understand various protocols for network security to protect against the threats in the networks.		

	SYLLABUS	**Contact Hours**
Module I	Introduction: Services, Mechanisms and attacks, the OSI security Architecture, Network security model, Classical Encryption techniques (Symmetric cipher model, substitution techniques, transposition techniques, steganography). Finite Fields And Number Theory : Groups, Rings, Fields Modular arithmetic, Euclid's algorithm, Finite fields, Polynomial Arithmetic, Prime numbers, Fermat's and Euler's theorem, The Chinese remainder theorem- Discrete logarithms.	**6 Hours**
Module II	Block Ciphers: Data Encryption Standard, Block cipher principles, block cipher modes of operation, Advanced Encryption Standard (AES) Public key cryptography: Principles of public key cryptosystems, The RSA algorithm, Key management, Hash Function and Digital Signatures: Authentication requirement, Authentication function, MAC, Hash function, Security of hash function and MAC, MD5, SHA, HMAC, CMAC, Digital signature and authentication Protocols.	**6 Hours**
Module III	Security Practice and System Security: Authentication applications, Kerberos, Internet Firewalls, Roles of Firewalls, Firewall related terminology, Firewall designs, Intruder, Intrusion detection system, Virus and related threats, Countermeasures, Firewalls design principles.	**6 Hours**
Module IV	E-mail Security: Security Services for Email attacks, establishing keys	**6 Hours**

	privacy, authentication of the source, Message Integrity.	
Module V	IP Security: Overview of IPSec, IP and IPv6, Authentication Header, Encapsulation Security Payload (ESP) Internet Key Exchange (Phases of IKE, ISAKMP/IKE Encoding). Web Security: SSL/TLS Basic Protocol, computing the keys- client authentication, PKI as deployed by SSL Attacks fixed in v3, Exportability, Encoding, Secure Electronic Transaction (SET).	**6 Hours**
Essential Reading	1. William Stallings, "Cryptography and Network security Principles and Practices", Pearson/PHI. 2. Wade Trappe, Lawrence C Washington, "Introduction to Cryptography with coding theory", Pearson.	
Supplementary Reading	1. W. Mao, "Modern Cryptography – Theory and Practice", Pearson Education. 2. Charles P. Pfleeger, Shari Lawrence Pfleeger – Security in computing – Prentice Hall of India.	
Course Outcomes:	**CO1:** Students will have the basic knowledge of different types of Security attacks. Symmetrical and Asymmetrical cryptography .To discuss on various types of attacks and their characteristics. **CO2:** To illustrate the basic concept of encryption and decryption for secure data transmission. Students will be able to analyze and compare different security mechanisms and services and Digital signature and authentication Protocols. **CO3:** Students will be able to analyze different modern encryption algorithms. Students will have the basic knowledge of different authentication Mechanisms and Security Practice and System Security **CO4:** Describe network security services and mechanisms. Security Services for Email attacks, establishing keys privacy, authentication of the source, Message Integrity **CO5:** Various network security applications, IPSec, Firewall, IDS, Web security, Email security, and malicious software etc.	

Object-Oriented Analysis and Design

Subject Code:		**Total Contact Hour**	**30**
Semester:	5th	**Total Credit**	**3**
Subject Name:	**Object-Oriented Analysis and Design**		
Pre-requisites (if any):			
Course Objectives:	1. To understand the fundamental concepts of object-oriented analysis and design (OOAD) and their significance in software development. 2. To learn various object-oriented methodologies and modelling techniques for system analysis and design. 3. To apply Unified Modelling Language (UML) diagrams for effective representation of software architecture and design. 4. To explore software design patterns and their role in creating reusable, maintainable, and scalable software solutions. 5. To develop problem-solving skills using object-oriented principles for real-world software applications.		
SYLLABUS			
Module I	**Fundamentals of Object-Oriented Concepts:** Object Modelling and New Paradigm, Object-Oriented Thinking and Rethinking, Objects and Classes, Links and Associations, Generalization and Specialization, Inheritance and Grouping Concepts, Aggregation and Composition, Abstract Classes and Polymorphism, Metadata and Constraints, Reuse,		**6 Hours**

	Object-Oriented Lifecycle Model, Overview of Object-Oriented Methodologies: OOD, HOOD, OMT, CRC, OOA, OOSA, OOSE, OOSD, OORASS	
Module II	**Architecture and Analysis:** Introduction to System Architecture, System Development as Model Building, Model Architecture, Requirements Model, Analysis Model, Design Model, Implementation Model, Test Model, Requirements Model Development, Analysis Model Development	**6 Hours**
Module III	**Construction and Testing:** Introduction to Construction, Design Model, Block Design, Working with Construction, Introduction to Testing: Unit Testing, Integration Testing, System Testing, The Testing Process	**6 Hours**
Module IV	**Modelling with UML:** Origin of UML, 4+1 View Architecture of UML, Basic Building Blocks of UML, Conceptual Model of UML, Basic Structural Modelling, UML Diagrams, Case Studies	**6 Hours**
Module V	**Advanced Object-Oriented Methodologies and Applications:** Advanced Object-Oriented Methodologies, Best Practices in Object-Oriented Design, Software Design Patterns, Component-Based Development, Emerging Trends in Object-Oriented System Design, Real-World Applications and Case Studies	**6 Hours**
Essential Reading	1. Grady Booch, "Object-Oriented Analysis and Design with Applications," Addison-Wesley 2. Craig Larman, "Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development," Pearson	
Supplementary Reading	1. Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, "Design Patterns: Elements of Reusable Object-Oriented Software," Addison-Wesley. 2. Bertrand Meyer, "Object-Oriented Software Construction," Prentice Hall 3. Martin Fowler, "UML Distilled: A Brief Guide to the Standard Object Modeling Language," Addison-Wesley	
Course Outcomes:	CO1: Demonstrate a clear understanding of object-oriented principles such as inheritance, polymorphism, encapsulation, and abstraction. CO2: Develop and interpret UML diagrams to represent different aspects of software systems. CO3: Analyse system requirements and design object-oriented models using appropriate methodologies. CO4: Implement software design patterns to enhance software reusability, maintainability, and efficiency. CO5: Evaluate and apply advanced object-oriented methodologies in the development of high-quality software applications.	

Web Technology

Subject Code:		**Total Contact Hour**	**30**
Semester:	**5th**	**Total Credit**	**3**
Subject Name:	**Web Technology**		
Pre-requisites (if any):	Basics of Programming in C		
Course Objectives:	1. To introduce the foundational concepts of the web, internet protocols, and the architecture of web applications. 2. To enable students to design HTML web page using basic HTML tags, CSS and forms. 3. To enable students to design dynamic HTML using JavaScript Language to perform functionalities at client side. 4. To provide practical knowledge about the development of graphical user interface applications. 5. To equip students with knowledge of web server for the management and delivery of information		

SYLLABUS		
Module I	Internet Architecture: Internet overview, evolution of internet. Internet components: Local Area Networks, Access Networks, Core Networks, Routers, Transmission infrastructure, ISPs. HTML: HTML Overview, Structure of HTML Documents, Document Types, HTML Elements and attributes. Anchor Attributes, Image Tag and its attributes, Image and Anchors, Table.	**6 Hr.**
Module II	Image Map: Attributes, Client-Side Image Maps and Server-Side Maps. HTML Layout: Background, colors and text, Tables, Frames, Layers, Page content Division<Div>, <SPAN>. CSS: Style Sheet Basic, Properties, Positioning with Style Sheet. Forms: Elements, Form controls. Dynamic HTML.	**6 Hr.**
Module III	Java Script: Introduction, Client-Side JavaScript, Server-Side JavaScript, JavaScript Objects, JavaScript Security. Operators: Assignment Operators, Comparison Operators, Arithmetic Operators, Increment, Decrement, Unary Negation, Logical Operators, String Operators, Special Operators, Conditional operator, Comma operator, delete, new, this, void. Control Statements.	**6 Hr.**
Module IV	JavaScript (Properties and Methods of Each): Array, Boolean, Date, Function, Math, Number, Object, String, regExp. Document and its associated objects, document, Link, Area, Anchor, Image, Applet, Layer. Events and Event Handlers: General Information about Events, Defining Event Handlers.	**6 Hr.**
Module V	Server-Side Programming: Introduction to Common Gateway Interface (CGI), Active Server Pages and PHP for accessing database. Internet applications: FTP, Telnet, Email, Chat. World Wide Web: HTTP protocol. Search Engines. E-commerce, Internet telephony, and virtual reality over the web, etc. Intranet and Extranet, firewall.	**6 Hr.**
Essential Reading	1. Web Design the Complete Reference, Thomas A. Powell, Tata McGraw-Hill. 2. Computer Networking: A Top-Down Approach Featuring the Internet, Kurose and Ross.6th Edition.	
Supplementary Reading	1.1. HTML The Complete Reference, Thomas Powell, Tata McGraw-Hill. 2. JavaScript the Complete Reference, Second Edition, Thomas Powell, Fritz Schneider. Tata McGraw-Hill.	
Course Outcomes:	CO1: Analyze internet architecture. CO2: Implement HTML web page using basic HTML tags, CSS, forms, dynamic HTML. CO3: Implement dynamic HTML using JavaScript Language to perform functionalities at client side. CO4: Develop graphical user interface applications for user interaction. CO5: Organize and implement web server for the management and delivery of information	

Sessional

Theory of Computation Lab

Subject Code:		**Total Contact Hour**	**20**
Semester:	5TH	**Total Credit**	**1.5**
Lab Name:	**Theory of Computation Lab**		
Course Objectives:	1. Understand the relevance and the concept of sustainability and the global initiatives in this direction 2. Explain the different types of environmental pollution problems and their sustainable solutions 3. Discuss the environmental regulations and standards 4. Outline the concepts related to conventional and non-conventional energy 5. Demonstrate the broad perspective of sustainable practices by utilizing engineering knowledge and principles		

Sl. No.	**Expt. No.**	**Experiment Details**
1	1	Design a DFA over the input set {a,b} accept all the string a. starting with symbol a. b. ending with symbol b. c. starting with symbol a and ending with symbol b
2	2	Design a Program for creating machine that accepts three consecutive one.
3	3	Design a Program for Mode 3 Machine.
4	4	Design a program for accepting binary number divisible by 2.
5	5	Design a program for creating a machine which accepts string having even no. of 1's and 0's.
6	6	Design a program for creating a machine which count number of 1's and 0's in a given string.
7	7	Design a Program to find 2's complement of a given binary number.
8	8	Design a Program which will increment the given binary number by 1.
9	9	Design a PDA to accept WCW^R where W is any string and W^R is reverse of that string and C is a Special symbol.
10	10	Design a Turing machine that's accepts the following language $a^nb^nc^n$ where n>0.

Course Outcomes:	CO1: Explain the basic concepts of switching and finite automata theory & languages. CO2: Relate practical problems to languages, automata, computability and complexity. CO3: Construct abstract models of computing and check their power to recognize the languages. CO4: Analyse the grammar, its types, simplification and normal form. CO5: Interpret rigorously formal mathematical methods to prove properties of languages, grammars and automata.

Operating Systems Lab

Subject Code:		**Total Contact Hour**	**20**
Semester:	5th	**Total Credit**	**1.5**
Lab Name:	**Operating Systems Lab**		
Course Objectives:	1. To learn operating system installation and basic command-line operations. 2. To implement process management, inter-process communication, and threading. 3. To develop and analyze CPU scheduling and deadlock handling algorithms. 4. To apply memory management and page replacement techniques. 5. To Design file system management and disk scheduling algorithms.		

Sl. No.	**Expt. No.**	**Experiment Details**
1	1	Installation of windows operating system.
	2	Illustrate UNIX commands and Shell Programming.
2	3	Process Management using System Calls: Fork, Exit, Getpid, Wait, Close
	4	Write C programs to implement the various CPU Scheduling Algorithms
3	5	Illustrate the inter process communication strategy
	6	Implement mutual exclusion by Semaphore
4	7	Write C programs to avoid Deadlock using Banker's Algorithm
	8	Write a C program to Implement Deadlock Detection Algorithm
5	9	Write C program to implement Threading
6	10	Implement the paging Technique using C program
	11	Write C programs to implement the following Memory Allocation Methods a. First Fit b. Worst Fit c. Best Fit
7	12	Write C programs to implement the various Page Replacement Algorithms
8	13	Write C programs to Implement the various File Organization Techniques
	14	Implement the following File Allocation Strategies using C programs a. Sequential b. Indexed c. Linked
9	15	Write C programs for the implementation of various disk scheduling algorithms
10	16	Install any guest operating system like Linux using VMware
Course Outcomes:		CO1: Demonstrate the installation of OS, use commands, and write shell scripts. CO2: Implement processes, system calls, and threads. CO3: Develop CPU scheduling and process synchronization algorithms. CO4: Apply memory management and page replacement strategies. CO5: Design file system management and disk scheduling algorithms

Artificial Intelligence and Machine Learning Lab

Subject Code:		**Total Contact Hour**	**20**
Semester:	5th	**Total Credit**	**1.5**
Lab Name:	**Artificial Intelligence and Machine Learning Lab**		
Course Objectives:	1. To understand and implement fundamental AI/ML algorithms and machine learning models. 2. To implement neural networks, including Multi-Layer Perceptron (MLP). 3. To explore and apply unsupervised learning techniques, such as k-Means. 4. To explore and apply unsupervised learning techniques, such as k-NN. 5. To explore and apply unsupervised/unsupervised learning techniques to real world datasets.		

Sl. No.	**Expt. No.**	**Experiment Details**
1	1	Implement Breadth-First-Search and find the path from the initial state to goal state
2	2	Implement Depth-First-Search and find the path from the initial state to goal state
3	3	Implement a Multi-Layer Perceptron with initial inputs, weights and output. Provide the final set of weights that will provide the corresponding output.
4	4	Implementation of Radical Basis Neural Network
5	5	Implement K-Means Algorithm
6	6	Implement K-Medoid
7	7	Implement Density based Clustering
8	8	Implement Decision Tree Algorithm
9	9	Implement Naive Bayes Classifier
10	10	Implement K- Nearest Neighbor (KNN) classification Algorithm

Course Outcomes:	CO1: Implement fundamental AI/ML algorithms. CO2: Develop and evaluate supervised and unsupervised learning models. CO3: Use appropriate ML evaluation metrics to analyze model performance. CO4: Design and train neural networks for classification and regression tasks. CO5: Apply AI/ML techniques to real-world datasets for better generalization.

Third Year (6th Semester)

Theory

Compiler Design

Subject Code:		**Total Contact Hour**	**30**
Semester:	**6TH**	**Total Credit**	3
Subject Name:	**Compiler Design**		
Pre-requisites (if any):	**Theory of Computation** **Computer Organization and architecture** **Computer Programming and Data Structures**		
Course Objectives:	1. Explain the phases of a Compiler 2. Illustrate the translation of regular expression into parse tree using syntax analyzer 3. Construct the intermediate representation considering the type systems 4. Apply the optimization techniques for the generated code 5. Use the different compiler construction tools to develop a simple compiler		

SYLLABUS		
Module I	Translator & its types, Brief introduction of the Compiler, Cousins of the Compiler, Analysis-synthesis model of compilation, Phases of a compiler, Front-end and Back-end of the compiler, Concept of Pass, Lexical analysis: Role of Lexical Analyzer, Input buffering, Specification & Recognition of Tokens, Design of a Lexical Analyzer Generator.	**6 Hours**
Module II	CFGs, Parsing Types, Top-down parsing, Brute force approach, recursive descent parsing, transformation on the grammars, predictive parsing, bottom-up parsing, operator precedence parsing, LR parsers (SLR, LALR, LR), Parser generation.	**6 Hours**
Module III	Syntax Directed Definitions, Evaluation Orders for Syntax Directed Definitions, Type Checking, storage organization, Storage allocation strategies, Symbol table, Error Detection & Recovery.	**6 Hours**
Module IV	Different Intermediate Forms - Abstract Syntax tree, Polish Notation, 3 address code, Implementation of 3 address code-Quadruple, Triples and Indirect triples, Code Generation: Issues in the design of code generator, Basic block and flow graphs, Register allocation, and assignment, DAG representation of basic blocks, code generation using DAG.	**6 Hours**
Module V	Classification of optimization, Principle sources of optimization, Optimization of basic blocks, Peephole Optimization, Loops in flow graphs, Local optimization, Global optimization, Data flow analysis of flow graph.	**6 Hours**
Essential Reading	1. Compilers: Principles, Techniques and Tools, Second Edition, Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffry D. Ullman. 2. Compiler Design in C By Allen I. Holub, Prentice-Hall/Pearson.	
Supplementary Reading	1. Lex & Yacc – John R. Levine, Tony Mason, Doug Brown, O'reilly 2. Compiler Construction, Louden, Thomson. 3. Advanced Compiler Design and Implementation By Muchnick, Morgan and Kaufmann	
Course Outcomes:	CO1: Design Lexical analyzer generator by using regular expressions and finite automata. CO2: Design and implement LL and LR parsers and use YACC Tool for developing a parser. CO3: Explain the applications of SDT and different types of intermediate-code generation CO4: Identify the storage organization used to support the run-time environment of	

	a program and effectively generate machine codes CO5: Apply the several algorithms for collecting and optimizing the information using data flow analysis

Software Engineering

Subject Code:		**Total Contact Hour**	**30**
Semester:	**6th**	**Total Credit**	**3**
Subject Name:	**Software Engineering**		
Pre-requisites (if any):	**Operating System, Computer Organization and Architecture, Data Structure and Algorithm, Programming Languages C, Python**		
Course Objectives:	1. Acquire knowledge of basic software engineering methods and practices 2. Understand the role of project management 3. Design solutions in one or more application domains using different design approaches 4. Acquire knowledge of various software testing strategies 5. Understand the various measures taken to ensure good quality software		
SYLLABUS			
Module I	**Introduction** - Evolution, Software Development Projects, Exploratory Style of Software Development. **Software Life Cycle Models**- Basic concepts, Waterfall Model and its extensions, Rapid Application Development, Spiral Model, Comparison of different life cycle models.		**06 Hours**
Module II	**Software Project Management**- Software Project Management Complexities, Responsibilities of a Software Project Manager, Project Planning, Metrices for project size estimation, Project Estimation Techniques, Empirical Estimation Techniques, COCOMO, Staffing Level Estimation, Scheduling, Organization and team structures, Staffing, Risk Management, Software Configuration Management. **Requirement Analysis & Specification**- Requirement Gathering, Requirement Analysis, Software Requirements Specification.		**06 Hours**
Module III	**Software Design**- Characteristics of good design, Cohesion and Coupling, Layered arrangement of modules. **Function-Oriented Software Design**- Structured Analysis, Developing DFD, Structured Design, Detailed Design, Design Review. **Object Modelling Using UML**- UML, UML Diagrams- Use Case, Class Diagram, Object Diagram, Sequence Diagram, Collaboration, Activity and State Chart.		**06 Hours**
Module IV	**Coding-** Standards & Guidelines, Code Review, Software Documentation **Testing-** Basic Concepts, Unit Testing, Black Box Testing, White Box Testing, Debugging, Integration Testing, System Testing		**06 Hours**
Module V	**Software Reliability and Quality Management**- Software Reliability, Statistical Testing, Software Quality, Software Quality Management System, ISO 9000, SEI Capability Maturity Model. **Computer Aided Software Engineering**- CASE Environment, CASE support in Software Life Cycle **Software Maintenance**- Characteristics of Software Evolution, Types of Software Maintenance, Software Reverse Engineering, Software Maintenance Process Models **Software Reuse**- Introduction, Issues in Reuse, Reuse Approach		**06 Hours**
Essential Reading	Rajib Mall, “Fundamental of Software Engineering”, 4th Edition, PHI		
Supplementary Reading	Roger S. Pressman, "Software Engineering: A practitioner's approach", McGraw Hill.		
Course Outcomes:	**CO1**: To understand the fundamental principles of software engineering, life cycle models and their appropriate applications. **CO2:**To understand the project management and requirement analysis techniques		

	CO3: To transform a specification into a design, and identify the components to build the architecture for a given problem using an appropriate methodology. **CO4:**To evaluate appropriate testing plans at different levels of software development **CO5:**To understand maintenance activities and quality control to ensure good quality software

Professional Elective - II	
Specialization-1	Evolutionary Computing
Specialization-2	Pattern Recognition
Specialization-3	Embedded Systems
Specialization-4	Cloud and Edge Computing
Specialization-5	Information Security and Management
Specialization-6	Software Testing and Quality Assurance
Specialization-7	Computer Graphics and Animation

Evolutionary Computing

Subject Code		**Total Contact Hour**	**30**
Semester	6th	**Total Credit**	**03**
Subject Name	**Evolutionary Computing**		
Pre-requisites(if any)	Basic Mathematical Knowledge		
Course Objective	To provide evolutionary Computation and global optimization techniques. To understand, analyze and apply the concepts of Neural network, neuro-modelling, several neural network paradigms. To understand, analyze and apply the concept of evolutionary computing paradigm known as genetic algorithm to engineering optimization problems. To understand, analyze and apply the concept of hybrid algorithms in different engineering application		

SYLLABUS		
Module I	**Introduction:** Evolutionary computing: Biological foundation of Evolutionary computing, Introduces evolutionary algorithms, a class of stochastic, population-based algorithms inspired by natural evolution theory, optimization problem solving.	04 **Hours**
Module II	**Neural Network:** Artificial Neuron and its model, activation functions, Neural network architecture: single layer and multilayer feed forward networks, recurrent networks, perception model, feed forward neural network, Back propagation, Adaline, Widrow-Hoff's Adaline model, Madaline, Unsupervised learning neural network: Hopfield neural network, Competitive learning, self-organizing feature map, Reinforcement learning: Q-learning, Temporal difference learning.	06 **Hours**
Module III	**Genetic Algorithm:** Fundamentals of genetic algorithms: Encoding, Fitness functions, Reproduction Genetic Modeling: Cross cover, Inversion and deletion, Mutation operator, Bit-wise operators, Bitwise operators used in GA. Convergence of Genetic algorithm. GA as an alternative to back propagation, Applications of GA in navigational planning of robots , Real life Problems	08 **Hours**
Module IV	Swarm Intelligence (SI): Biological foundation of SI, SI Techniques: Ant	08

	Colony Optimization (ACO) and Particle Swarm optimization (PSO). General steps in ACO, the "Invisible Manager"(Stigmergy), the Pheromone, Ant Colonies and Optimization, Ant Colonies and Clustering, Applications of Ant Colony Optimization. Applications of ACOPSO: Social Network Structure: The Neighborhood Principle, PSO Algorithm, Fitness Calculation, Convergence, PSO System Parameters, Particle Swarm Optimization versus Evolutionary Computing and Applications of PSO	**Hours**
Module V	**Hybrid Systems:** Neuro-GA, Adaptation of neural learning algorithm using GA, Mimetic algorithm, Firefly Algorithm, multi objective algorithms	04 **Hours**
Essential Reading	Computational Intelligence Principles, Techniques and Applications, Amit Konar, Springer publication. M. Mitchell, An introduction to Genetic Algorithms, Prentice-Hall D. E. Goldberg, "Genetic Algorithms in Search, Optimization, and Machine Learning, "Addison Wesley R. Eberhart, P. Simpson and R. Dobbins, Computational Intelligence - PC Tools, AP Professional	
Supplementary Reading	Neuro Fuzzy and Soft Computing, J. S. R. JANG,C.T. Sun, E. Mitzutani, PHI Soft-computing Fundamentals and Applications D. K. Pratihar, Alpha Science	
Course Outcomes	CO1: Describe the Evolutionary algorithms and solve complex problem using evolutionary algorithms CO2: To solve the problems using neural network techniques CO3: Identify the issues in design and implementation of genetic algorithm CO4: Explain the concepts of Swarm Intelligence techniques CO5: Understand the concept of Hybrid Systems and its objectives.	

Pattern Recognition

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Pattern Recognition**		
Pre-requisites (if any):			
Course Objectives:	Students will understand basic concepts in pattern recognition and to Explain and compare a variety of pattern classification, structural pattern recognition, and pattern classifier combination techniques. Analyze classification problems probabilistically and estimate classifier performance. Understand the principles of Bayesian parameter estimation and apply them in relatively simple probabilistic models.		
SYLLABUS			
Module I	Introduction: Machine Perception, Pattern Recognition Systems, Design Cycle, Learning and Adaptation, Training and Learning in Pattern Recognition Approach, Applications of Pattern Recognition, Patterns and Features, Different Types of Pattern Recognition.		**6 Hours**
Module II	Probability: Introduction, Probability of Events, Random Variables, Joint Distributions and Densities, Moments of Random Variables, Estimation of Parameters from Samples, Minimum Risk Estimators.		**6 Hours**

Module III	Statistical decision making: introduction, baye's theorem, multiple features, conditionally independent features, decision boundaries, unequal costs of error, estimation of error rates, the leaving one- out technique. Characteristic curves, estimating the composition of populations.	**6 Hours**
Module IV	Nonparametric decision making: introduction, histograms, kernel and window estimators, nearest neighbor classification techniques, adaptive decision boundaries, adaptive discriminate functions, minimum squared error discriminate functions, choosing a decision-making technique.	**6 Hours**
Module V	Unsupervised Learning and Clusterings: Unsupervised Bayesian Learning, Data Decryption and Clustering, Criterion Functions and Clustering, Hierarchical Clustering, Online Clustering, Component Analysis.	**6 Hours**
Essential Reading	Pattern Classification Duda R. O., Hart P E., Stork D G., Wiley Publishers Pattern Recognition and Image Analysis, Earl Gose, Richard J and Steve J, PHI	
Supplementary Reading	Pattern Recognition (Statistical, Structural and Neural Approaches), Robert Schalkoff.	
Course Outcomes:	CO1: Understand basic concepts in pattern recognition and to Explain and compare a variety of pattern classification, structural pattern recognition, and pattern classifier combination techniques. CO2: Analyze classification problems probabilistically and estimate classifier performance. CO3: Understand the principles of Bayesian parameter estimation and apply them in relatively simple probabilistic models. CO4: Learn decision making techniques and different function. CO5: Gain knowledge about different types of clustering technique and its implementation in real life problems.	

Embedded Systems

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**03**
Subject Name:	**Embedded Systems**		
Pre-requisites (if any):	**Microprocessor and Microcontroller** **Operating Systems**		
Course Objectives:	1. Acquire knowledge about embedded system design and ARM processor. 2. Study different devices, device drivers and some Interfacing devices. 3. Obtain knowledge on various Real Time Operating systems along with scheduling algorithms. 4. Understand the requirement of Hardware Software partitioning and Hardware Software co-simulations. 5. Know about low power devices and its characteristics for embedded system design.		
SYLLABUS			
Module I	Introduction: Features of Embedded systems, Design matrices, Embedded system design flow. ARM**:** An advanced Micro Controller, Brief history, ARM pipeline, Instruction Set Architecture (ISA): Registers, ARM Instructions, THUMB, FPGA.		**06 Hours**
Module II	Interfacing: Devices and device drivers, Serial peripheral interfaces (SPI), Inter Integrated Circuit (IIC), RS232C, RS422, RS485, USB, Infrared communication (IrDA), Bluetooth.		**06 Hours**
Module III	Real time operating systems: Hard real time, Firm real time, Soft real		**08**

	time, Task periodicity: Periodic task, Sporadic task, Aperiodic task, Task scheduling, Scheduling algorithms: Clock driven scheduling, Event driven scheduling, Commercial Real-time operating systems.	**Hours**
Module IV	Hardware-software partitioning and Co-simulation: K-L partitioning, Hardware-software co-simulation: Dimensions in co-simulation, Co-simulation approaches, Emulation, Abstract level co-simulation, Detailed level co-simulation.	**05 Hours**
Module V	Low power embedded system design: Dynamic power dissipation, Static power dissipation, Power reduction techniques, System level power management, Introduction to NoC.	**05 Hours**
Essential Reading	1. S. Chattopadhyay, "Embedded System Design", PHI. 2. Raj Kamal, "Embedded system architecture, programming and design", TMH.	
Supplementary Reading	1. Ralf Niemann, "Hardware software co-design of Embedded systems", Kulwer Academic. 2. Sriram V Iyer, Pankaj Gupta, "Embedded real time system programming", TMH. 3. Rajib Mall, "Real-Time Systems-Theory and practice", Pearson Education 4. Peter Marwedel, "Embedded System Design", Springer 5. John Catsoulis, "Designing Embedded Hardware", O'Reilly	
Course Outcomes:	CO1: Acquire knowledge about embedded system design and ARM processor. CO2: Understand and select different Interfacing devices for embedded system design. CO3: Analyze and apply various Real Time Operating systems along with scheduling algorithms for efficient embedded system design. CO4: Evaluate for selection of efficient Hardware Software partitioning algorithm for a particular embedded system design. CO5: Able to design low power devices for embedded systems.	

Cloud and Edge Computing

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Cloud and Edge Computing**		
Pre-requisites (if any):	1. Basic understanding of computer networks and operating systems. 2. Familiarity with fundamental programming concepts.		
Course Objectives:	1. To introduce the foundational concepts, characteristics, and models of Cloud Computing. 2. To explore the underlying infrastructure and virtualization technologies that enable cloud environments. 3. To familiarize students with common cloud services and technologies offered by major providers. 4. To cover essential aspects of cloud security and management, including cost optimization and migration. 5. To provide a basic understanding of Edge Computing principles, architecture, and applications.		
SYLLABUS			
Module I	Fundamentals of Cloud Computing: Inception to Future Trends, Introduction to Cloud Computing, Cloud Service Models, Cloud Deployment Models, Case Studies		**06 Hours**
Module II	Cloud Infrastructure and Virtualization: Data Centers and Cloud Infrastructure, Virtualization, Cloud Storage Concepts, Case Studies		**06 Hours**
Module III	Cloud Services and Technologies: Networking in the Cloud, Cloud Databases, Cloud Compute Services, Case Studies		**06 Hours**
Module IV	Cloud Security and Management: Cloud Security Fundamentals, Cloud		**06**

	Cost Management, Cloud Migration Strategy, Case Studies	**Hours**
Module V	Introduction to Edge Computing" Need, Advantages, Goals, Fundamentals of Edge Computing, Edge Devices and Architecture, Use Cases and Applications of Edge Computing, Case Studies	**06 Hours**
Essential Reading	1."Cloud Computing: A Hands-on Approach", Arshdeep Bahga and Vijay Madisetti 2."Cloud Computing: Principles and Paradigms", Rajkumar Buyya, James Broberg, and Andrzej Goscinski	
Supplementary Reading	1."Distributed Systems: Concepts and Design" by George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair 2. "Edge Computing 101: Expert Techniques And Practical Applications", Rob Botwright 3. "Cloud Computing", Shailendra Singh	
Course Outcomes:	CO1: Understand the fundamental concepts of Cloud Computing, CO2: Comprehend the essential components of cloud infrastructure, with a focus on the role of virtualization technologies & different types of cloud storage. CO3: Identify and describe the core services offered by cloud providers for computing, networking, and databases. CO4: Recognize foundational aspects of cloud security, including identity and access management, and strategies for cloud cost management. CO5: Grasp the core concepts of Edge Computing, identify various edge devices and architectural components, and recognize applications domains of Edge Computing.	

Information Security and Management

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Information Security and Management**		
Pre-requisites (if any):	Basic understanding of operating systems, database systems, networking concepts and basic security principles.		
Course Objectives:	4. To know the fundamental concepts on information security and an introduction to cryptography. 5. To understand the methods for authentication and access control mechanisms. 6. To know about viruses, malicious code and security in operating systems. 7. To acquire understanding of the concepts of security in networks and database systems. 8. To analyze a structured approach to deal with security incidents when they occur, legal and ethical issues in information security.		
SYLLABUS			
Module I	**Information Security Basics:** What is Computer Security? Security Goals: Confidentiality, Integrity, Availability (CIA Triad), Threats and Vulnerabilities, Best Practices and Principles for Information Security. **Introduction to Cryptography:** Plain Text and Cipher Text, Encryption and Decryption, Types of Cryptography, Basics of Encryption Algorithms in Information Security, Digital Signatures.		**8 Hours**
Module II	**Authentication and Access Control:** Authentication principle and its working, Authentication Methods, Authentication vs. Authorization, Access control and its working, Types of Access Control, Access Control Policies.		**5 Hours**
Module III	**Viruses and Malicious Code:** Secure Programs, Non-malicious Program		**7**

	Errors, Viruses and other Malicious Code, Countermeasures against Program Threats. **Operating Systems Security:** Security in Operating Systems, Threats to operating system, Operating System Security Measures - Authorization, User Authentication, Access Control, File Protection, Security Policies.	**Hours**
Module IV	**Security in Networks:** Threats in Network, Wireless Network Security, Denial of Service, Security Countermeasures - Cryptography in Network Security, Firewalls, Intrusion Detection and Prevention Systems. **Database Security:** Security requirements of database, Reliability and integrity, Sensitive Data, Types of Attacks.	**6 Hours**
Module V	**Information Security Incident Management:** Goals, **Components of an** Information Security Incident Management System**, Incident Response Plan (IRP). Legal and Ethical Issues:** Protecting Programs and Data, Information and the Law, Employees Rights, Ethical Issues in Computer Security.	**4 Hours**
Essential Reading	3. Security in Computing, Fifth Edition, by Charles P. Pfleeger, Pearson Education 4. Network Security Essentials: Applications and Standards, by William Stallings. Prentice Hall.	
Supplementary Reading	1. B. A. Forouzan & D Mukhopadhyay, Cryptography and Network Security. McGraw Hill. 2. Cryptography and Network Security: Principles and Practice, William Stallings, Pearson Publication.	
Course Outcomes:	CO1: Be familiar with security goals, attacks, and introductory concepts of cryptography. CO2: Learn the various authentication and access control mechanisms. CO3: Gain knowledge in malicious codes, viruses and security in operating systems. CO4: Analyze security in networks and database systems. CO5: Learn different security incident management policies as well as legal and ethical issues relating to information security.	

Software Testing and Quality Assurance

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Software Testing and Quality Assurance**		
Pre-requisites (if any):	**Software Engineering**		
Course Objectives:	1. Understand fundamental concepts of software testing and quality assurance. 2. Learn various software testing techniques and methodologies. 3.Understand test management and defect tracking in SDLC 4. Explore software quality models and assurance practices. 5. Gain hands-on experience with testing tools and automation.		
SYLLABUS			
Module I	**Fundamentals of Software Testing**: Introduction to basics of software testing, its importance in the software development life cycle, Verification and Validation. Types of software testing, functional and non-functional testing, software testing principles.		**6 Hours**
Module II	**Testing Techniques and Strategies**: Black-box testing techniques:		**6**

	equivalence partitioning and boundary value analysis, White-box testing techniques: control flow, data flow, statement, branch, and path coverage. Various testing levels: unit, integration, system, and acceptance testing. Regression testing, Smoke testing, and Agile and DevOps testing	**Hours**
Module III	**Test Management and Defect Tracking**: Test planning, strategy formulation, and documentation. Effective test cases writing, Defect Life Cycle and Bug Tracking (JIRA, Bugzilla), Test metrics and measurements, Risk-based testing.	**6 Hours**
Module IV	**Software Quality Assurance (SQA) and Automation**: Overview of software quality assurance, differentiating it from quality control. Software quality models: McCall's, Boehm's, ISO 9001, and CMMI. Automated testing concepts and tools including Selenium, JUnit, TestNG, and LoadRunner.	**6 Hours**
Module V	**Advanced and Emerging Trends in Testing**: Advanced testing methodologies including performance testing (load, stress, scalability), security testing, and penetration testing. Cloud-based testing approaches, AI and machine learning in software testing. Mobile application testing.	**6 Hours**
Essential Reading	1.Naresh Chauhan: Software Testing Principles and Practices, Second Edition, Oxford University Press 2.RenuRajani, Pradeep Oak: Software Testing, Effective Methods, Tools and Techniques, Second Edition, Mc Graw Hill	
Supplementary Reading	1. Ammann, P., & Offutt, J. - Introduction to Software Testing (2nd Edition), Cambridge University Press. 2. Pressman, R. S., & Maxim, B. R. - Software Engineering: A Practitioner's Approach (9th Edition), McGraw-Hill. 3. Kan, S. H. - Metrics and Models in Software Quality Engineering (2nd Edition), Addison-Wesley.	
Course Outcomes:	CO1: Solve the problems using Software Testing techniques and Approaches. CO2: Apply various Software testing Techniques to find bugs in software. CO3: Use open-source software Testing Tools. CO4: Apply various Software Quality Assurance Techniques to ensure the quality in software. CO5: Comparison of different testing tools with suitable cases	

Computer Graphics and Animation

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Computer Graphics and Animation**		
Pre-requisites (if any):	Mathematics: Linear algebra (vectors, matrices), calculus, and geometry are essential for transformations, shading, and rendering. Physics: Basic understanding of physics concepts like motion, lighting, and optics helps in realistic animation and rendering. Programming: Proficiency in languages like C++, Python, or Java is crucial, especially for graphics programming. Basic Computer Science: Understanding of algorithms, data structures, and memory management.		
Course Objectives:	1. To understand the basics of various input and output devices used in computer graphics. 2. To learn the basic to build 2D and 3D objects. 3. To understand 2D and 3D transformations and Viewing. 4. To implement project on objects. 5. To be familiar with Animation and surface detection methodologies.		
SYLLABUS			
Module I	Introduction to Computer Graphics- Overview of Graphics System,		**7**

	Mathematical Foundations and Applications. Graphics System- Cathode Ray Tube, Random Scan and Raster Scan Systems, Colour CRT Monitors. Output Primitives – Scan Conversion, Points & Lines, Line Drawing Algorithms (DDA Algorithm And Bresenham‘S Line Algorithm). Circle Generation Algorithm- Properties of Circles, Midpoint Circle Algorithm. Ellipse-Generation Algorithm- Properties of Ellipses, Midpoint Ellipse Algorithm. Polygons Types and Filling Algorithm- Types of polygon (Convex and Concave), Scan Line Polygon Fill Algorithm, Boundary Fill Algorithm and Flood Fill Algorithm. Anti-Aliasing.	**Hours**
Module II	Two-Dimensional Geometric Transformations- Basic Transformations, Matrix Representations & Homogeneous Coordinates, Composite Transformations, Other Transformations. Two-Dimensional Viewing – The Viewing Pipeline, Viewing Coordinate Reference Frame, Widow-To- Viewport Coordinate Transformation. Clipping Operations – Point Clipping, Line Clipping (Cohen Sutherland) and Polygon Clipping (Sutherland Hodgman) Algorithms.	**6 Hours**
Module III	Three-Dimensional Geometric Transformation- Basic Transformations, Matrix Representations & Homogeneous Coordinates, Other Transformations. Projections- Types of Projections (Parallel and Prospective), Mathematical Description of Projections. Three-Dimensional Object Representation- Curve Design, Blending Function and Continuity, Spline Curve, Bezier Curves and Surfaces. Fractal Geometry Methods- Fractal Generation Procedure, Classification of Fractals Dimension.	**6 Hours**
Module IV	Visible Surface Detection- Hidden Lines and Surfaces, Back Face Detection, Depth Comparison, Z Buffer Algorithm, A Buffer Algorithm. Illumination Model- Basic Illumination Models, Displaying Light Intensities, Halftone Pattern and Dithering Techniques. Surface Rending Methods: Polygon Rendering Methods, Gouraud and Phong Shading.	**6 Hours**
Module V	Computer Animation – Raster Methods for Computer Animation, Design of Animation Sequences, Traditional Animation Techniques, General Computer-Animation Functions, Computer-Animation Languages, Key-Frame Systems, Motion Specifications, Character Animation, Periodic Motions, OpenGL Animation Procedures.	**5 Hours**
Essential Reading	1. Hearn, D., Baker, M. P., & Carithers, W. R. - Computer Graphics with OpenGL (4th ed.). Pearson Education. 2. Foley, J. D., van Dam, A., Feiner, S. K., Hughes, J. F., & Akeley, K. Computer Graphics: Principles and Practice (3rd ed.). Addison-Wesley.	
Supplementary Reading	1. Hearn, D., & Baker, M. P. - Computer Graphics, C Version (2nd ed.). Pearson Education. 2. Parent, R. - Computer Animation (3rd ed.). Morgan Kaufmann.	
Course Outcomes:	CO1: Acquire familiarity with the relevant mathematics of computer graphics. CO2: Be able to design basic graphics application programs, including animation CO3: Be able to design applications that display graphic images to given specifications. CO4: Assess the importance of viewing and projections. CO5: Understand the fundamentals of animation, virtual reality and its related technologies.	

Professional Elective - III	
Specialization-1	Deep Learning
Specialization-2	Big data Analysis
Specialization-3	Advanced Computer Architecture
Specialization-4	Internet of Things
Specialization-5	Digital Forensics
Specialization-6	E-Commerce and ERP
Specialization-7	Simulation and Modeling

Deep Learning

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Deep Learning**		
Pre-requisites (if any):	Vectors, matrices, matrix multiplication, eigen values, eigenvectors, singular value decomposition, Probability distributions, conditional probability, Bayes' theorem, expectation, variance, common distributions (normal, binomial, Poisson), Differentiation, integration, partial derivatives, chain rule, gradients, Jacobian and Hessian matrices, Gradient descent, convex functions, Lagrange multipliers, constrained optimization. ,Machine Learning: Supervised learning, unsupervised learning, classification, regression, clustering, overfitting, underfitting, model evaluation metrics., Artificial Neural Networks		
Course Objectives:	1. Understand the Base and Fundamentals of Deep Learning 2. Understand deep learning architectures 3. Explore Convolution neural networks 4. Understand Sequence Models 5. Explore Unsupervised Neural Networks and its advancement		
SYLLABUS			
Module I	Artificial Neural Network (ANN) Basics: ANN architecture, Learning algorithms, Multilayer Perceptron, Back-propagation algorithm and its variants Stochastic gradient decent, Curse of Dimensionality, Need of Deep Learning. Deep learning models.		**5 Hours**
Module II	Deep Learning Architectures: Machine Learning vs. Deep learning, Representation Learning, Width and Depth of Neural Networks, Activation Functions: Sigmoid, Tanh, ReLU & Leaky ReLU, SoftMax.		**6 Hours**
Module III	Convolutional Neural Networks: Architectural Overview, Motivation, Layers, Filters, Parameter sharing, Regularization, Popular CNN Architectures: ResNet, Alexnet –Applications.		**6 Hours**
Module IV	Sequence Models. Recurrent Neural Networks, Backpropagation over time. LSTM, GRU. BPTT for training RNN, Long Short Term Memory Networks. Language Models. DL applications.		**7 Hours**
Module V	Unsupervised Neural Networks, Restricted Boltzmann Machines. Motivation for Sampling, Markov Chains, Training RBMs - Gibbs Sampling, Contrastive Divergence. Autoencoder, Generative Adversarial Networks (GANs)		**6 Hours**
Essential Reading	1. “Deep Learning”, MIT Press book, by Ian Goodfellow and Yoshua Bengio and Aaron Courville 2. “Dive into deep learning”. Cambridge University Press, by Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J.		

Supplementary Reading	1. Deep Learning: Foundations and Concepts, Springer, by Christopher M. Bishop and Hugh Bishop 2. Understanding Deep Learning, The MIT Press, by Simon J.D. Prince
Course Outcomes:	CO1: Comprehend Deep Learning Principles CO2: Develop and Implement Neural Networks CO3: Implement Deep Learning Models CO4: Skills in Model Optimization, Model Selection and Performance Analysis CO5: Apply Deep Learning Solutions to Real-World Problems

Big Data Analysis

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Big Data Analysis**		
Pre-requisites (if any):			
Course Objectives:	1. To Introduce basic concepts of Big Data Systems. 2. To Provide students with knowledge and abilities to develop applications using Hadoop, Hive, Apache Pig, etc. 3. Introduce to MongoDB 4. Introduce to Data Analysis with Spark and Core Spark Concepts. 5. To provide students with knowledge on NoSQL Databases.		

SYLLABUS

Module I	What is Big Data and where it is produced? Rise of Big Data, Compare Hadoop vs traditional systems, Limitations and Solutions of existing Data Analytics Architecture, Attributes of Big Data, data warehouse v/s Data Lakes, other technologies vs Big Data. What is Hadoop? Hadoop History, Distributing Processing System, Core Components, HDFS Architecture, Hadoop Master – Slave Architecture, Daemon types - Learn Name node, Data node, Secondary Name node.	**8 Hours**
Module II	What is Hadoop Cluster? Pseudo Distributed mode, Type of clusters, Hadoop Ecosystem, Pig, Hive, Oozie, Flume, SQOOP. Overview of Map Reduce Framework, Map Reduce Architecture. Learn about Job tracker and Task tracker, Use cases of Map Reduce, Anatomy of Map Reduce Programs.	**6 Hours**
Module III	Hive DDL – Create/Show/Drop Tables, Hive DML – Load Files & Insert Data, Hive Architecture & Components, Partitions in Hive, PIG Architecture & Data types, Shell and Utility components, PIG Latin: File Loaders and UDF, Programming structure in UDF, PIG, limitations of PIG. NoSQL Databases: Base Architecture, HBase Components, Storage Model of HBase, Object Data Stores-S3, HBase vs RDBMS. Introduction to MongoDB, CRUD, Advantages of MongoDB over RDBMS, Use case.	**6 Hours**
Module IV	Introduction to Data Analysis with Spark, Core Spark Concepts, Programming with RDDs, RDD Basics, Creating RDDs, RDD Operations, Passing Functions to Spark. Accumulators, Broadcast Variables, working on a Per-Partition Basis, piping to External Programs, Numeric RDD Operations Spark Runtime Architecture, Deploying Applications with spark-submit.	**5 Hours**
Module V	Using Spark SQL in Applications, Loading and Saving Data from RDDs, JDBC/ODBC Server, User-Defined Functions. Why Mahout, Data too large for single machine Data already on Hadoop, Algorithms implemented in Mahout, Setting up the development environment, Mahout API, Parallel versus in-memory execution mode.	**5 Hours**
Essential Reading	1. BIG DATA and ANALYTICS, Seema Acharya, Subhashini Chellappan, Wiley publications.	

	2. Hadoop: The Definitive Guide, 4th Edition. Publisher: O'Reilly Media, Inc. Release
Supplementary Reading	Student's Handbook for Associate Analytics-III, NASSCOM.
Course Outcomes:	CO1:Understand Big Data concepts, including cloud and big data architectures. CO2:Motivate and explain trade-offs in big data processing technique design and analysis. CO3:Apply non-relational databases, the techniques for storing and processing large volumes of structured and unstructured data, as well as streaming data. CO4:Integrate Apache Mahout with newer platforms such as Apache Spark. CO5:Understand and analyze OLAP & OLTP processing with Map Reduce

Advanced Computer Architecture

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Advanced Computer Architecture**		
Pre-requisites (if any):	**Computer Architecture/Organization**		
Course Objectives:	1. The objective of this course is to analyze the parallelism, identify the conditions of Parallelism and study different parallel interconnection systems. 2. It also focuses on Identifying the pipeline hazards, gain in-depth knowledge of architecture and learn Parallel processing and its applications to solve workloads		

SYLLABUS		
Module I	Flynn's classification: SISD, SIMD, MISD, MIMD, Parallel Processing: Definition, Theory of Parallelism. Parallel Computer Models, Parallelism in Uni-processor computers, Implicit Parallelism vs. Explicit Parallelism, Levels of parallelism. Software Parallelism, Hardware Parallelism, Amdahl's law, Overview of RISC and CISC architecture, System Performance attributes of parallel Computers.	**6 Hours**
Module II	Pipelining: Linear pipe line processor, Asynchronous and Synchronous models, Speed up, Efficiency, Throughput, Instruction pipeline, Arithmetic pipeline, Pipelining in MIPS architecture, Nonlinear pipe line processor, Conditions of Parallelism: Data and Resource Dependencies, Control Dependence, Bernstein's condition, Pipeline Hazards and their Resolution Mechanisms. Super scalar, Super pipeline and VLIW architectures.	**8 Hours**
Module III	Loosely coupled and tightly coupled system, Parallel Interconnection Systems: Static and Dynamic Networks, Single stage Interconnection Network, Multistage Interconnection Networks, Control Structure, Node degree, Diameter, Bisection width, Network Latency, Bandwidth, Scalability, Data routing functions:- Permutation, Perfect shuffle exchange, Hypercube Routing function.	**6 Hours**
Module IV	SIMD architecture, Matrix multiplication through SIMD computers, Array Processor, Vector Processor.	**4 Hours**
Module V	Memory hierarchy, Cache Design Issues, Memory Interleaving, Symmetric and Distributed memory architecture, Case studies of some recent commercial processors.	**6 Hours**
Essential Reading	1. J. L Hennessy and D. A. Patterson, "Computer Architecture – A quantitative approach", Elsevier 2. Kai Hwang , "Advanced Computer Architecture", TMH	
Supplementary Reading	1. I Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar, "Introduction to Parallel Computing", Pearson Education	

	2. Kai Hwang & F. Briggs , "Advanced Computer Architecture", MGH 3. J. L Hennessy and D. A. Patterson, "Computer Organization & Design", Elsevier
Course Outcomes:	CO1: Understand and remember the concepts of parallelism in computers to identify the conditions for achieving parallelism. CO2: Analyze pipeline concepts and different pipeline hazards and able to apply hazard resolution techniques to enhance system performance. CO3: Study different Interconnection Networks for better data communication. CO4: Evaluate different types of parallel processing for a particular application to enhance system performance. CO5: Analyzing concepts of different memory access techniques to design performance-aware memory.

Internet of Things

Subject Code:		**Total Contact Hour**	**30**
Semester:	**6th**	**Total Credit**	**3**
Subject Name:	**Internet of Things**		
Pre-requisites (if any):	**Computer Network, Internet and Web Technology, Computer Architecture**		
Course Objectives:	1. To know the basic concepts of IoT. 2. To study about Access technologies in IoT 3. To study the design methodology and different IoT hardware platforms. 4. To study the basics of Data Analytics and supporting services used in IoT. 5. To study about various applications of IoT in real world and industrial applications.		

SYLLABUS		
Module I	Overview of Internet of Things: IoT-An Architectural Overview–Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. M2M and IoT Technology Fundamentals- Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service(XaaS), M2M and IoT Analytics, Knowledge Management	**6 Hours**
Module II	IoT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.11ah and Lora WAN, Network Layer: IP versions, Constrained Nodes and Constrained Networks,6LoWPAN, Application Transport Methods: SCADA, Application Layer Protocols: CoAP and MQTT.	**6 Hours**
Module III	Design Methodology, Embedded computing logic, Microcontroller, System on Chips, IoT system building blocks IoT Platform overview: Overview of IoT supported Hardware platforms such as: Raspberry pi, Arduino Board details, Security in IoT Protocols	**6 Hours**
Module IV	Data Analytics And Supporting Services: Data Analytics: Introduction, Structured Versus Unstructured Data, Data in Motion versus Data at Rest, IoT Data Analytics Challenges, Data Acquiring, Organizing in IoT/M2M, Supporting Services: Computing Using a Cloud Platform for IoT/M2M Applications/Services, Everything as a service and Cloud Service Models.	**6 Hours**

Module V	Case Study and IoT Application Development: IoT applications in home-infrastructures security – Industries – IoT electronic equipments. Use of Big Data and Visualization in IoT Industry 4.0 concepts	**6 Hours**
Essential Reading	1. IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry, Cisco Press, 2017. 2. Internet of Things – A hands-on approach, Arshdeep Bahga, Vijay Madisetti, Universities Press, 2015 3. Internet of Things: Architecture, Design Principles And Applications, Raj kamal, McGraw Hill Higher Education 4. Peter Waher, "Learning Internet of Things", PACKT publishing, BIRMINGHAM – MUMBAI	
Supplementary Reading	1. The Internet of Things – Key applications and Protocols, Olivier Hersent, David Boswarthick, Omar Elloumi and Wiley. (for Unit2). 2. "From Machine-to-Machine to the Internet of Things-Introduction to a New Age of Intelligence", Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stamatis Karnouskos, Stefan Aves and. David Boyle, Elsevier. 3. Architecting the Internet of Things, Dieter Uckelmann, Mark Harrison, Michahelles and Florian (Eds), Springer.	
Course Outcomes:	CO1: Understand the basics of IoT CO2: Implement the state of the Architecture of IoT CO3: Understand design methodology, protocols used, hardware platforms, and design constraints involved in IoT CO4: Understand how to analyze and organize the data. CO5: Study and Compare real time IOT Applications in Industrial & world.	

Digital Forensic

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Digital Forensic**		
Pre-requisites (if any):			
Course Objectives:	1. Understand the fundamentals of incident response, cyber security forensics principles, and their relevance to cybersecurity operations. 2. Develop proficiency in preparation, including the formulation of policies, incident handling workflows, and the use of various incident response tools. 3. Gain expertise in the identification phase by mastering techniques for detection, triage, and incident classification, along with the use of indicators of compromise (IOCs). 4. Acquire the skills needed for effective containment, including damage limitation, system isolation, and forensic backup and imaging, while limiting malware spread. 5. Explore the digital forensics investigation process, including applicable laws, evidence collection, chain of custody, and the use of technical forensics tools and techniques, such as those for analyzing hard disks, file systems, network devices, and mobile devices.		
SYLLABUS			
Module I	Introduction: Definitions of incident response and forensic analysis, relation of incident response to the rest of cyber security operations, incident response phases -preparation, identification, containment, eradication, recovery, follow-up, indicators of compromise (IOC), forensic analysis as an incident response tool and as support for cybercrime investigations, cyber security forensics principles.		**6 Hours**
Module II	Preparation, Identification, Containment		**6**

	Preparation: Policies and procedures, incident workflows, guidelines, incident handling forms, principles of malware analysis, log analysis, threat intelligence, vulnerability management, penetration testing, digital forensics, incident ticketing systems, incident documentation templates. Identification: Detection, incident triage, information gathering and reporting, incident classification, indicators of compromise (IOC). Containment: Damage limitation, network segment isolation, system isolation, forensic backup and imaging, use of write blockers, temporary fixes, malware spread limitation.	**Hours**
Module III	Eradication, Recovery, Follow-up Eradication: Actual removal and restoration of affected systems, removal of attack artifacts, scanning of other systems to ensure complete eradication, use of IOCs on other systems and local networks, cooperation with forensic analysis to understand the attack fully. Recovery: Test and validate systems before putting back into production, monitoring of system behavior, ensuring that another incident will not be created by the recovery process. Follow-up: Documenting lessons learned preparatory activities for similar future incidents, technical training, and process improvement.	**6 Hours**
Module IV	Digital Forensics Investigation Process: Applicable laws, investigation methodology, chain of custody, evidence collection, digital evidence principles, rules and examination process, first responder procedures.	**6 Hours**
Module V	Technical forensics tools and techniques: Hard disks, removable media and file systems, Windows forensics, duplication/imaging of forensic data, recovering deleted files and hidden or deleted partitions, steganography and image forensics, log analysis, password crackers, network device forensics, packet capture analysis, email tracking, mobile forensics, investigation of attacks, common tools (Encase, FTK, etc.)	**6 Hours**
Essential Reading	1.“Incident Response & Computer Forensics, Third Edition” by Jason T. Luttgens and Matthew Pepe 2. “Blue Team Handbook: Incident Response Edition: A condensed field guide for the Cyber Security Incident Responder”, by Don Murdoch2.	
Supplementary Reading	1.Computer Incident Response and Forensics Team Management: Conducting a Successful Incident Response”, by Leighton Johnson 2. The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics”, by John Sammons5. “Digital Forensics with Open Source Tools”, by Cory Altheide and Harlan Carvey 3. Digital Forensics Processing and Procedures”, by David Lilburn Watson and Andrew Jones	
Course Outcomes:	CO1: Apply incident response phases, policies, and procedures in real world cyber security scenarios. CO2: Effectively identify and classify security incidents using indicators of compromise (IOCs) and triage techniques. CO3: Demonstrate proficiency in containing and mitigating security incidents while limiting damage and malware spread CO4: Conduct digital forensics investigations in compliance with applicable laws and chain of custody requirements. CO5: Utilize a range of technical forensics tools and techniques to analyze digital evidence and investigate cyber attacks.	

E-Commerce & ERP

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**03**
Subject Name:	**E-Commerce & ERP**		

Pre-requisites (if any):		
Course Objectives:	1. To understand the concept, importance, role, and various activities of E-Commerce. 2. To understand the different infrastructure required to developed e-Commerce site. 3. To demonstrate the concepts of security in e-commerce applications. 4. To understand the concept of digital payment systems, E-Marketing and its tools in E-Commerce. 5. To understand the different modules of ERP Software packages and their uses in in industries.	
SYLLABUS		
Module I	**Introduction:** E-Commerce vs. Physical Commerce, Different Types of E-commerce with examples, Perspectives, Scenarios, Examples of E-Commerce, Advantages, Disadvantages of E-Commerce and Overview to INTERNET .**WWW:** Basic network architecture, layered model, Next Generation Internet, Web system architecture, URL, **HTTP:** request, response, Generation of dynamic web pages, Cookies.	**6 hours**
Module II	**Client side programming**: Important Factors in web programming, Web page Design and Production, **HTML:** Overview, Structure, Text Formatting, Links, Tables, Images, Frames, Form, Java Script, **Server side programming:** 3-Tier Model, CGI, ASP, Java servlet, **Database Connectivity:** Relational database System, JDBC Perspectives, and brief overview to Session Tracking.	**6 hours**
Module III	**Basic cryptography for enabling e-commerce:** Security concerns and requirements, Encryption (Private & Public key), RSA, Stream and block Cipher, Message Digest, Digital Signature and its Standard, Authentication, internet **security:** IPsec protocol, Firewalls, Different types of Firewalls, Advance technologies for **e-commerce:** Introduction to mobile agents, Brief overview to WAP, XML, Data Mining.	**6 hours**
Module IV	**Internet payment systems:** Characteristics of payment system, 4‘C payment method, SET Protocol, digital signature generation and verification, e-Cash, e-Check, Micropayment Systems**Consumer oriented e-commerce:** Traditional vs. e-Retailing and their key success factors; e-Retailing - Benefits, Features and Models, **Business oriented ecommerce:** Features of B2B e-Commerce, Business Models, **E-services:** Categories, Web enabled Services, Matchmaking Services, e-Entertainment.	**7 hours**
Module V	**Enterprise Resource Planning (ERP):** Features, capabilities and Overview of Commercial Software, re-engineering work processes for IT applications, Business Process Redesign, Knowledge Engineering and Data Warehouse. Business Modules: Finance, Manufacturing (Production), Human Resources, Plant Maintenance, Materials, Management, Quality Management Sales & Distribution ERP Package. ERP Market: ERP Market Place, SAP AG, People Soft, BAAN, JDEdwards, Oracle Corporation.	**5 hours**
Essential Reading	1.E-Commerce Fundamentals & Applications by Henry Chan, Raymond Lee, Tharam Dillon, John Wiley & Sons, Ltd. 2. E-commerce - An Indian Perspective by P.T. Joseph, S.J , PHI publication	
Supplementary Reading	1. E-Commerce, MM Oka, EPH 2. Kalakotia, Whinston: Frontiers of Electronic Commerce, Pearson Education. 3. Bhaskar Bharat: Electronic Commerce – Technologies & Applications, TMH 4. Loshin Pete, Murphy P.A.: Electronic Commerce, Jaico Publishing Housing 5. Enterprise Resource Planning: Fundamentals of Design and Implementation 6. By Ganesh, K., Mohapatra, S., Anbuudayasankar, S.P., Sivakumar, P., Springer	
Course Outcomes:	CO1. Analyze the foundations and importance of E-commerce and basic network architecture of internet, WWW. CO2. Demonstrate of client side and server side programming tools for building E-	

	Commerce applications CO3. Demonstrate Basic cryptography, internet security and advance technologies for enabling e-commerce. CO4. Analyze the impact and use of Internet payment systems, Consumer oriented e-commerce, and Business oriented e-commerce and E-services. CO5. Evaluate and discuss global E-commerce issues, characterize current trends in ERP system development and investigate how e-commerce has affected the way people buy goods and services.

Simulation and Modeling

Subject Code:		**Total Contact Hour**	**30**
Semester:	6th	**Total Credit**	**3**
Subject Name:	**Simulation and Modeling**		
Pre-requisites (if any):			
Course Objectives:	1. To learn different types of systems, models and simulation techniques, their importance, application, processes etc. 2. To study various types of random number generation techniques, random variates generation techniques, testing of random numbers, application of random numbers in simulation. 3. To study discrete event simulation and select correct language for system simulation and be able to write codes for various applications. 4. To design and evaluate simulation experiments, perform different types of validations, study and apply different variance reduction techniques to real life problems.		

SYLLABUS		**Contact Hours**
Module I	**Introduction:** Introduction to Simulation and system models, major application areas, concept of a System, Environment, Continuous and Discrete systems, System Modelling, Types of Models, Progress of a Simulation Study, Monte Carlo Method, Comparison of Simulation and Analytical Methods, Numerical Computation Technique for discrete and continuous models	**6 Hours**
Module II	**Random Number Generation and Testing :** Random Number Generators: Linear Congruential Generator, Additive and Multiplicative Congruential Generators, Mid Square Method, Rejection Method, Testing of random numbers: Kolmogorov Smirnov test, Chi-Square test, Runs test, Generation of Stochastic Variates using Inverse Transformation technique, Acceptance rejection technique	**8 Hours**
Module III	**Discrete System Simulation:** Discrete Events, Representation of Time, Generation of Arrival Patterns, Fixed Time Step versus Next Event Simulation Model, Simulation of a Telephone System.	**4 Hours**
Module IV	**Simulation Languages and Practical Systems:** Continuous and Discrete Systems Languages, Factors in the selection of Discrete Systems Simulation Language, Computer Model of different systems like, Queuing, Inventory etc **General Purpose System Simulation (GPSS):** Creating and moving transactions, Queues, Facilities and Storages, Gathering Statistics, Conditional Transfers, Program Control statements, Priorities and Parameters, Standard Numerical Attributes, Functions, Gates, Logic Switches and Tests, Variables, Select and Count, Example GPSS problems	**8 Hours**
Module V	**Design and Evaluation of Simulation Experiments:** Length of Simulation Runs, Validation, Variance Reduction Techniques	**4 Hours**
Essential Reading	1. Geoffrey Gordon, “System Simulation”, PHI Publication 2. Narsingh Deo, “System Simulation with Digital computer”, PHI Publication	

	3. Jerry Banks and John S.Carson, Barry L. Nelson, David M.Nicol, "Discrete Event System Simulation", 3rd Edition, Prentice Hall Publication.
Supplementary Reading	1. Shannon, R.E, “Systems simulation, The art and science”, Prentice Hall Publication. 2. Thomas J. Schriber, “Simulation using GPSS”, John Wiley Publication.
Course Outcomes:	**CO1:** To know the importance of modelling and simulation, types of models, system and the reflection of it’s behaviour in models. **CO2:** To learn the techniques of random number generation and testing and its applications in simulation. **CO3:** To analyze discrete system simulation. **CO4:** To build computer models of Queuing, Inventory, Scheduling systems etc. **CO5:** To enable students choose appropriate simulation languages and be able to code using them properly.

Sessional

Compiler Design Lab

Subject Code:		**Total Contact Hour**	**20**
Semester:	6th	**Total Credit**	**1.5**
Lab Name:	**Compiler Design Lab**		
Course Objectives:	1. To understand the various phases in the design of a compiler 2. To understand the design of top-down and bottom-up parsers. 3. To understand syntax directed translation schemes. 4. To understand code generation and code optimization algorithm. 5. To introduce lex and yacc tools.		
Sl. No.	**Expt. No.**	**Experiment Details**	
1	1	Write a C program to identify different types of Tokens in a given Program.	
2	2	Write a Lex Program to implement a Lexical Analyzer using Lex tool.	
3	3	Write a C program to Simulate Lexical Analyzer to validating a given input String.	
4	4	Write a C program to implement the Brute force technique of Top down Parsing.	
5	5	Write a C program to implement a Recursive Descent Parser.	
6	6	Write C program to compute the First and Follow Sets for the given Grammar.	
7	7	Write a C program for eliminating the left recursion and left factoring of a given grammar	
8	8	Write a C program to check the validity of input string using Predictive Parser.	
9	9	Write a C program for implementation of LR parsing algorithm to accept a given input string.	
10	10	Write a C program for implementation of a Shift Reduce Parser using Stack Data Structure to accept a given input string of a given grammar.	
11	11	Simulate the calculator using LEX and YACC tool.	
12	12	Generate YACC specification for a few syntactic categories.	
13	13	Write a C program for generating the three address code of a given expression /statement.	
14	14	Write a C program for implementation of a Code Generation Algorithm of a given expression/statement.	
Course Outcomes:	CO1: Understand the working of lex and yacc compiler for debugging of programs. CO2: Understand and define the role of lexical analyzer, use of regular		

	expression and transition diagrams. CO3: Understand and use Context free grammar, and parse tree construction. CO4: Develop program for solving parser problems. CO5: Learn how to write programs that execute faster.

Software Engineering Lab

Subject Code:		**Total Contact Hour**	**20**
Semester:	6th	**Total Credit**	**1.5**
Lab Name:	**Software Engineering Lab**		
Course Objectives:	1. Hands-on experience in applying key software engineering principles, such as requirements gathering, system design, coding, testing, and deployment, through practical lab exercises and projects. 2. Modern software development methodologies and tools to manage and execute software projects effectively. 3. Improve problem-solving and analytical skills 4. Emphasize the importance of clear, concise, and comprehensive documentation to ensure long-term success in software projects 5. Gain Exposure to Software Development Tools and Technologies		

Sl. No.	**Expt. No.**	**Experiment Details**
1	1	To write down the problem statement for any application development project or for any suggested system of relevance.
2	2	To perform requirement analysis and develop Software Requirement Specification (SRS) document for suggested system.
3	3	To develop the function-oriented diagram: Data Flow Diagram (DFD) and Structural Chart.
4	4	To perform the user's view analysis for the suggested system: Use case diagram.
5	5	To draw the structural view diagram for the system: Class diagram, object diagram.
6	6	To draw the behavioral view diagram: State-chart diagram, Activity diagram.
7	7	To perform the behavioral view diagram for the suggested system: Sequence diagram, Collaboration diagram.
8	8	To perform the implementation view diagram for the suggested system: Sequence diagram, Collaboration diagram.
9	9	To perform the environmental view diagram: Deployment diagram for the system.
10	10	To perform various testing operations using the testing tool unit testing, integration testing for a suggested system.
11	11	To perform Estimation of effort using FP Estimation for chosen system.
12	12	To prepare time line chart/Gantt chart/PERT chart for selected software project.
Course Outcomes:		**CO1** Implement the various open-source tools could be applied in different specific domain. **CO2:** Use the software tools and methodology to develop the software product. **CO3:** Make the software requirement specification document of any given problem statement. **CO4:** Draw and develop the UML diagram based on different views i.e., users view, behavioral view, implementation view etc. **CO5:** Perform the time line for the PERT chart and GANT chart for the given software projects metrics.